6.1 Pythagorean Theorem

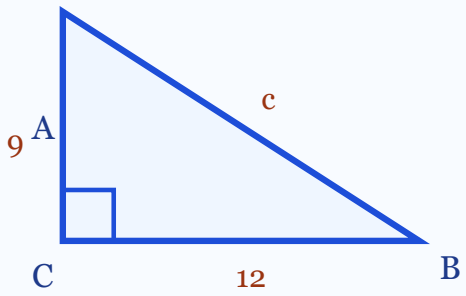
In a right triangle with legs a and b and hypotenuse c, the Pythagorean Theorem says $a^2 + b^2 = c^2$. The converse lets you classify any triangle as acute, right, or obtuse by comparing $a^2 + b^2$ with c^2 (where c is the longest side). This worksheet goes beyond one-step triples: expect radicals, multi-step applications, and classification with careful ordering of sides.

Examples

Example 1: Find the hypotenuse (full steps)

In right $\triangle ABC$ with right angle at C, $AC = 9$ and $BC = 12$. Find AB.

Solution



Right angle at C; hypotenuse is opposite the right angle.

Identify the hypotenuse: $AB = c$ is opposite $\angle C$.
Apply $a^2 + b^2 = c^2$:
$9^2 + 12^2 = c^2$
$81 + 144 = c^2$
$c^2 = 225$
$c = \sqrt{225} = 15$ (length is positive).
Answer: $AB = 15$.

Example 2: Find a leg — leave exact form

A right triangle has hypotenuse 10 and one leg 4. Find the other leg in simplest radical form.

Solution

Let the unknown leg be x. Then

$4^2 + x^2 = 10^2$

$16 + x^2 = 100$

$x^2 = 84$

$x = \sqrt{84} = \sqrt{4 \cdot 21} = 2\sqrt{21}$.

Answer: $2\sqrt{21}$.

Example 3: Converse — prove it is right

A triangle has sides 20, 21, and 29. Show it is a right triangle.

Solution

Longest side is 29, so test whether $20^2 + 21^2 = 29^2$.

$20^2 + 21^2 = 400 + 441 = 841$

$29^2 = 841$

Equal ⇒ by the converse of the Pythagorean Theorem, the triangle is **right** (right angle opposite 29).

Example 4: Classify carefully

Sides 7, 10, and 12. Classify as acute, right, or obtuse.

Solution

Longest side $c = 12$. Compare $7^2 + 10^2$ with 12^2:

$49 + 100 = 149$ and 144.

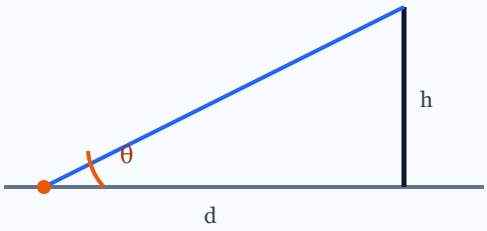
Since $a^2 + b^2 > c^2$ ($149 > 144$), the triangle is **acute**.

Memory: $>$ acute, $=$ right, $<$ obtuse (always using the longest side as c).

Example 5: Ladder against a wall

A 25 ft ladder leans against a wall. Its base is 7 ft from the wall. How high up the wall does it reach?

Solution



Model: ladder = hypotenuse; wall height and ground distance are legs.

Let h be the height on the wall.

$7^2 + h^2 = 25^2$

$49 + h^2 = 625$

$h^2 = 576$

$h = 24$ ft.

Check: 7-24-25 is a scaled 7-24-25 triple. ✓

Example 6: Diagonal of a rectangle

A rectangular courtyard is 16 m by 30 m. Find the length of a diagonal walkway.

Solution

The diagonal is the hypotenuse of a right triangle with legs 16 and 30:

$d^2 = 16^2 + 30^2 = 256 + 900 = 1156$

$d = \sqrt{1156} = 34$ m.

Answer: 34 m.

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Example 7: Two steps — find then classify

A triangle has sides x, 15, and 17, and you are told $x^2 + 15^2 = 17^2$. Find x, then state what kind of triangle it is.

Solution

From the given: $x^2 + 225 = 289 \Rightarrow x^2 = 64 \Rightarrow x = 8$.

Sides 8, 15, 17 satisfy $8^2 + 15^2 = 17^2$, so the triangle is **right**.

Example 8: Isosceles right triangle

An isosceles right triangle has legs of length $5\sqrt{2}$. Find the hypotenuse.

Solution

Both legs equal $5\sqrt{2}$:

$c^2 = (5\sqrt{2})^2 + (5\sqrt{2})^2 = 50 + 50 = 100$

$c = 10$.

(Shortcut: in a 45-45-90 triangle, hypotenuse = leg $\cdot \sqrt{2} = (5\sqrt{2})\sqrt{2} = 10$.)

Example 9: Word problem with a missing piece

A guy wire is anchored 9 m from the base of a vertical pole and reaches a point 12 m up the pole. A second wire from the same anchor reaches the top of the pole. If the pole is 20 m tall, how long is the second wire?

Solution

The second wire is the hypotenuse for legs 9 and 20:

$L^2 = 9^2 + 20^2 = 81 + 400 = 481$

$L = \sqrt{481}$ m ($481 = 13 \cdot 37$, so this does not simplify further).

Answer: $\sqrt{481}$ m.

Note: The 9-12 wire is a red herring for the final length — draw the diagram and use the full height 20.

Example 10: Oblique classification trap

Sides 9, 40, and 41. A student claims $9^2 + 40^2 \neq 41^2$ so it is not right. Check their claim.

Solution

Compute carefully: $81 + 1600 = 1681$ and $41^2 = 1681$. They **are** equal — it is a right triangle (9-40-41 triple). The student likely mis-multiplied 41^2.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Hypotenuse

Legs 11 and 60. Find the hypotenuse.

Answer: **Verification:**

Problem 2: Leg (radical)

Hypotenuse 14, one leg 6. Find the other leg in simplest radical form.

Answer: **Verification:**

Problem 3: Converse

Sides 12, 35, 37. Is the triangle right? Show the test.

Answer: **Verification:**

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Problem 4: Classify

Sides 6, 9, 11. Acute, right, or obtuse?

Answer: **Verification:**

Problem 5: Ladder

A 29 ft ladder's base is 20 ft from a wall. How high does it reach?

Answer: **Verification:**

Problem 6: Diagonal

TV screen 27 in by 36 in (rectangular). Find the diagonal.

Answer: **Verification:**

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Problem 7: Classify

Sides 13, 14, 15. Classify.

Answer: **Verification:**

Problem 8: Exact length

Legs 5 and $5\sqrt{3}$. Find the hypotenuse.

Answer: **Verification:**

Problem 9: Missing leg

Hypotenuse $2\sqrt{13}$, leg 4. Find the other leg.

Answer: **Verification:**

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Problem 10: Application

A baseball diamond is a square with side 90 ft. Find the distance from home to second (exact form).

Answer: **Verification:**

Problem 11: Converse false friend

Sides 5, 12, 14. Right triangle? Justify.

Answer: **Verification:**

Problem 12: Multi-step

Right triangle: one leg is 8, hypotenuse is 3 more than the other leg. Find all three sides.

Answer: **Verification:**

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Problem 13: Perimeter path

A rectangular park is 40 m by 42 m. You walk the two short sides and one long side, or you cut the diagonal then walk one short side. Which route is shorter, and by how much?

Answer: **Verification:**

Problem 14: Challenge

An isosceles triangle has sides 10, 10, and base 12. Classify the apex angle (between the equal sides). Hint: compare $10^2 + 10^2$ with 12^2.

Answer: **Verification:**

Problem 15: Challenge

A wire from the top of a 15 m pole to the ground is 25 m long. How far from the base is the anchor? If the wire slips so the anchor moves 8 m farther from the base, how high on the pole does the wire now reach?

Answer: **Verification:**

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✓ Answer Key

1. 61
2. $4\sqrt{10}$
3. Yes; $12^2 + 35^2 = 37^2$
4. Obtuse ($36 + 81 = 117 < 121$)
5. 21 ft
6. 45 in
7. Acute ($169 + 196 = 365 > 225$)
8. 10
9. 6
10. $90\sqrt{2}$ ft
11. No; $25 + 144 = 169 \neq 196$ (obtuse)
12. Legs 8 and 15; hypotenuse 17
13. Diagonal+short is shorter by 14 m
14. Acute ($100 + 100 = 200 > 144$)
15. First distance 20 m; new height 7 m

Great work — keep practising!